Generalized Tensor Regression for Hyperspectral Image Classification

Jianjun Liu, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Liang Xiao, *Member, IEEE*, Jun Sun, and Hong Yan, *Fellow, IEEE*

***Abstract*— In this article, we propose a novel tensorial approach, namely, generalized tensor regression, for hyperspectral image classification. First, a simple and effective classifier, i.e., the ridge regression for multivariate labels, is extended to its tensorial version by taking advantages of tensorial representation. Then, the discrimination information of different modes is exploited to further strengthen the capacity of the model. Moreover, the model can be simplified and solved easily. Different from traditional tensorial methods, the proposed model can be utilized to capture not only the intrinsic structure of data in a physical sense but also the generalized relationship of data in a logical sense. Our proposed approach is shown to be effective for different classification purposes on a series of instantiations. Specifically, our experiment results with hyperspectral images collected by the airborne visible/infrared imaging spectrometer, the reflective optics spectrographic imaging system and the ITRES CASI-1500 demonstrate the effectiveness of the proposed approach as compared to other tensor-based classifiers and multiple kernel learning methods.**

***Index Terms*— Column generation (CG), hyperspectral image classification, multiple kernel/feature learning, tensor decomposition, tensor regression.**

I. Introduction

REMOTELY sensed hyperspectral image is a 3-D digital image captured by imaging sensors in hundreds of narrow and contiguous spectral bands spanning the visible to infrared spectrum. Each hyperspectral pixel can be treated as a vector comprising various spectral-band responses. The wealth of spectral information in hyperspectral image promotes its various applications, such as material identification. Among the tasks of material identification, supervised classification is an important one and has attracted many researchers' attention [1]–[3], where pixels are assigned to one of the available classes according to a set of given training pixels. Generally, the methods proposed for pattern recognition can be exploited for hyperspectral image classification, including support vector machines [4], multinomial logistic regression [5], sparse and collaborative representation [6]–[8], multiple kernel learning [9], deep learning [3], [10], and several other methods [11]–[13].

Different from the general problems of pattern recognition, hyperspectral pixels are an ordered list of spectral signatures, and they comply with a predefined spatial arrangement of image. That is to say, we should handle a hyperspectral pixel together with its neighboring ones. In the last decade, many research studies focus on the fusion of spatial and spectral information to improve the classification performance of hyperspectral images. For instance, there are some approaches that develop spatial–spectral features for the subsequent classification, such as morphological profiles [14], extinction profiles [15], [16], and deep features [17]–[19]. Some other approaches consider the fusion of spatial and spectral information during the classification process, such as sparse and collaborative representation models [20], [21], multiple kernel/feature methods [9], [22], [23], and deep models [3], [10]. Apart from these, there are some approaches that incorporate the spatial information by refining classification maps [8], [24], [25].

For the fusion of spatial and spectral information, traditional approaches mainly perform fixed feature extraction patterns in spatial domain, which can lead to the loss of intrinsic structure information for hyperspectral images. Inspired by the tensor learning methods, some approaches consider solving this problem by treating pixels and their neighboring pixels as 3-D tensors, and thereby propose tensorial classification methods. In [26], a multidimensional filter that performs both spatial filtering and spectral dimensionality reduction is proposed for the improvement of classification accuracy. Guo *et al.* [27] first use a multilinear principal component analysis to preprocess tensors and then employ support tensor machines for the final classification. In [28], two tensorial classification methods are proposed for hyperspectral data analysis, i.e., logistic tensor regression (LTR) and rank-1 feedforward

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J. Liu is with the Jiangsu Provincial Engineering Laboratory for Pattern Recognition and Computational Intelligence, Jiangnan University, Wuxi 214122, China, also with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong, and also with the Jiangsu Key Laboratory of Spectral Imaging & Intelligent Sense, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: liuofficial@163.com).

Z. Wu and L. Xiao are with the Jiangsu Key Laboratory of Spectral Imaging & Intelligent Sense, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zebin.wu@gmail.com; xiaoliang@mail.njust.edu.cn).

J. Sun is with the Jiangsu Provincial Engineering Laboratory for Pattern Recognition and Computational Intelligence, Jiangnan University, Wuxi 214122, China (e-mail: junsun@jiangnan.edu.cn).

H. Yan is with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong (e-mail: h.yan@cityu.edu.hk).

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neural network. Several other classification methods have been proposed in the literature by taking advantages of tensor representation [29]–[31]. Moreover, tensor representation has been successfully applied in many other applications of remotely sensed images, such as superresolution [32], [33], denoising [34], [35], anomaly detection [36], and spectral unmixing [37], [38]. Note that most of these tensorial methods build tensors physically and neglect the features that follow a predefined logical arrangement. It is necessary to construct generalized tensors by using these features and thereby capture the generalized relationship of them.

In this article, we propose a generalized tensor regression (GTR) approach for the general classification of hyperspectral images. The proposed approach is extended from a simple and efficient classifier, i.e., the ridge regression (RR) for multivariate labels [39], and leverages the structural information of tensors by incorporating the CANDECOMP/PARAFAC (CP) decomposition [40]. Furthermore, the prior knowledge of different modes, i.e., the nonnegativity, is exploited to enhance the discrimination of the model. Some of the innovative characteristics of the proposed approach are highlighted as follows.

1) First, the prior knowledge of different modes is considered for the improvement of the model, and the nonnegativity is exploited to promote the sparsity of GTR. For traditional tensorial methods, such as support tensor machines [27], [41], LTR [28], and rank regression [42], [43], they only devote to capturing the structural information by using tensorial techniques.
2) Second, GTR couples all classes and trains them together. This characteristic is different from the other methods of the rank regression that GTR belongs to. For instance, the approaches proposed in [42] and [43] are a binary classifier. In [43], the multiple classification is achieved by training each class separately.
3) Third, GTR is a generalized classifier that can be utilized to deal with not only the tensors built in a physical sense but also the generalized tensors constructed in a logical sense. Six kinds of tensors taken from different views are investigated to show the effectiveness of GTR.
4) Finally, GTR is a simple and stable classifier. The RR model can be seen as a special case of GTR, and the algorithm of GTR can be simplified by a divide-and-conquer strategy and directly solved by alternating update. Moreover, GTR is not sensitive to the initial values.

The remainder of this article is organized as follows. Section II briefly introduces some notations and preliminaries. In Section III, we first present the proposed GTR approach and the corresponding algorithms and then introduce six instantiations of tensorial features for evaluation. The effectiveness of the proposed approach is demonstrated in Section IV by conducting experiments on three hyperspectral scenes. Finally, Section V concludes this article with some remarks and future research directions.

II. Notations and Preliminaries

A. Tensor and Multilinear Algebra

A tensor is a multidimensional array, and its order is the number of dimensions, also known as modes [40]. Without loss of generality, vectors (i.e., first-order tensors) are denoted by boldface lowercase letters, e.g., $\mathbf{a}$, and matrices (i.e., second-order tensors) are denoted by boldface capital letters, e.g., $\mathbf{A}$. The ith row and jth column of $\mathbf{A}$ are denoted by $\mathbf{a}_{i:}$ and $\mathbf{a}_{:j}$, respectively. For higher order tensors, i.e., tensors of order three or higher, they are denoted by calligraphic letters, e.g., $\mathcal{A}$.

Given an Nth-order tensor $\mathcal{A} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$, its element is denoted as $a_{l_1 l_2 \ldots l_N}$ $(1 \leq l_n \leq L_n)$. The mode-n fibers of tensor $\mathcal{A}$ are the L_n-dimensional vectors $\mathbf{a}_{l_1 \ldots l_{n-1} : l_{n+1} \ldots l_N}$ extracted from $\mathcal{A}$ by fixing every index but l_n. The mode-n unfolding matrix $\mathbf{A}_{(n)} \in \mathbb{R}^{L_n \times L_1 \ldots L_{n-1} L_{n+1} \ldots L_N}$ is defined by arranging the mode-n fibers of $\mathcal{A}$ to be the columns of the resulting matrix. The inner product of two same-sized tensors $\mathcal{A}, \mathcal{W} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$ is the sum of the products of their entries, that is,

$$\langle \mathcal{A}, \mathcal{W} \rangle = \sum_{l_1=1}^{L_1} \sum_{l_2=1}^{L_2} \cdots \sum_{l_N=1}^{L_N} a_{l_1 l_2 \ldots l_N} w_{l_1 l_2 \ldots l_N}. \tag{1}$$

Then, for the Frobenius norm of $\mathcal{A}$, it is defined as $\|\mathcal{A}\|_F^2 = \langle \mathcal{A}, \mathcal{A} \rangle$. The outer product of a tensor $\mathcal{A} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$ and another tensor $\mathcal{A}' \in \mathbb{R}^{L'_1 \times L'_2 \times \cdots \times L'_M}$ is defined by

$$(\mathcal{A} \circ \mathcal{A}')_{l_1 l_2 \ldots l_N l'_1 l'_2 \ldots l'_M} = a_{l_1 l_2 \ldots l_N} a'_{l'_1 l'_2 \ldots l'_M}. \tag{2}$$

Definition 1 (The n-Mode Product): The n-mode product of a tensor $\mathcal{A} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$ with a matrix $\mathbf{W} \in \mathbb{R}^{I_n \times L_n}$, denoted by $\mathcal{A} \times_n \mathbf{W}$, is an Nth-order tensor $\mathcal{B} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times I_n \times \cdots \times L_N}$, whose elements are computed by

$$b_{l_1 \ldots l_{n-1} i_n l_{n+1} \ldots l_N} = \sum_{l_n=1}^{L_n} a_{l_1 \ldots l_{n-1} l_n l_{n+1} \ldots l_N} w_{i_n l_n}. \tag{3}$$

In Definition 1, each mode-n fiber is multiplied by the matrix $\mathbf{W}$, and thus the n-mode product can also be expressed in terms of matrix multiplication $\mathbf{B}_{(n)} = \mathbf{W}\mathbf{A}_{(n)}$. Given a series of matrices $\{\mathbf{W}^{(n)} \in \mathbb{R}^{I_n \times L_n} | n = 1, 2, \ldots, N\}$, we define a tensor $\mathcal{G} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ as

$$\mathcal{G} = \mathcal{A} \times_1 \mathbf{W}^{(1)} \times_2 \mathbf{W}^{(2)} \cdots \times_N \mathbf{W}^{(N)} = \mathcal{A} \prod_{n=1}^{N} \times_n \mathbf{W}^{(n)}. \tag{4}$$

Then, for any $n \in \{1, 2, \ldots, N\}$, we have

$$\mathbf{G}_{(n)} = \mathbf{W}^{(n)} \mathbf{A}_{(n)} \left(\prod_{i=N, i \neq n}^{1} \otimes \mathbf{W}^{(i)} \right)^T \tag{5}$$

where $\prod_{i=N, i \neq n}^{1} \otimes \mathbf{W}^{(i)} = \mathbf{W}^{(N)} \otimes \cdots \otimes \mathbf{W}^{(n+1)} \otimes \mathbf{W}^{(n-1)} \otimes \cdots \otimes \mathbf{W}^{(1)}$ and the symbol $\otimes$ denotes the Kronecker product.

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For distinct modes in a series of multiplications, the order of the multiplication is irrelevant, that is,

$$\mathcal{A} \times_1 \mathbf{W}^{(1)} \times_2 \mathbf{W}^{(2)} = \mathcal{A} \times_2 \mathbf{W}^{(2)} \times_1 \mathbf{W}^{(1)}. \tag{6}$$

Definition 2 (Rank-R CP Decomposition): Given a tensor $\mathcal{W} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$, the CP decomposition factorizes it into a sum of component rank-one tensors $\{\mathcal{W}_r \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N} | r = 1, 2, \ldots, R\}$, written as

$$\mathcal{W} = \sum_{r=1}^{R} \mathcal{W}_r = \sum_{r=1}^{R} \mathbf{w}_r^{(1)} \circ \mathbf{w}_r^{(2)} \circ \cdots \circ \mathbf{w}_r^{(N)} = \sum_{r=1}^{R} \prod_{n=1}^{N} \circ \mathbf{w}_r^{(n)} \tag{7}$$

where $\mathbf{w}_r^{(n)} \in \mathbb{R}^{L_n}$.

B. RR for Multivariate Labels

In this section, we introduce multiclass RR for hyperspectral image classification. Given a hyperspectral scene, suppose that it includes C classes and J training samples $\mathbf{A} \in \mathbb{R}^{L \times J}$ with L being the number of bands. For each training sample $\mathbf{a}_{:j} \in \mathbb{R}^L$, if it belongs to the cth class ($c \in \{1, 2, \ldots, C\}$), its label vector is defined by $\mathbf{t}_{:j} = [0, \ldots, 1, \ldots, 0]^T \in \mathbb{R}^C$ with only the cth element equal to one. The goal of RR is to estimate coefficient weights $\mathbf{W} \in \mathbb{R}^{C \times L}$ that can model the linear dependence between $\mathbf{a}_{:j}$ and $\mathbf{t}_{:j}$, which is achieved by solving the following optimization problem:

$$\min_{\mathbf{W}} \ \|\mathbf{T} - \mathbf{W}\mathbf{A}\|_F^2 + \lambda \|\mathbf{W}\|_F^2 \tag{8}$$

where the first term is the fidelity term, the second term is the regularization term, $\mathbf{T} = [\mathbf{t}_{:1}, \ldots, \mathbf{t}_{:J}] \in \mathbb{R}^{C \times J}$, and $\lambda > 0$ is a regularization parameter.

Problem (8) is convex and smooth, and its closed-form solution can be written as

$$\mathbf{W} = \mathbf{T}\mathbf{A}^T(\mathbf{A}\mathbf{A}^T + \lambda \mathbf{I})^{-1} \tag{9}$$

where $\mathbf{I}$ denotes the identity matrix. Then, we can determine the label of a test sample $\mathbf{x} \in \mathbb{R}^L$ by

$$y = \arg \max_{c=1,2,\ldots,C} \ \delta_c(\mathbf{W}\mathbf{x}) \tag{10}$$

where $\delta_c(\cdot)$ represents the cth element of a vector.

III. Proposed Approach

A. GTR

Before deriving our GTR model, we reformulate the optimization problem (8) as

$$\min_{\mathbf{w}_{c:}} \ \sum_{c,j} (t_{cj} - \langle \mathbf{a}_{:j}, \mathbf{w}_{c:} \rangle)^2 + \lambda \sum_c \|\mathbf{w}_{c:}\|_2^2. \tag{11}$$

In the above problem (11), each training sample $\mathbf{a}_{:j}$ exits as a vector, i.e., the input features are arranged as 1-D vectors. For hyperspectral image classification, such as spatial–spectral classification, the input features follow a predefined arrangement. It is necessary to maintain this structure during the classification process [27], [28]. We extend (11) from the vector space to the tensor space as

$$\min_{\mathcal{W}_{c:}} \ \sum_{c,j} (t_{cj} - \langle \mathcal{A}_{:j}, \mathcal{W}_{c:} \rangle)^2 + \lambda \sum_c \|\mathcal{W}_{c:}\|_F^2 \tag{12}$$

where $\mathcal{A}_{:j} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$ contains the input features of the jth training sample and $\mathcal{W}_{c:}$ is the weight tensor of equal number of orders and dimensions to $\mathcal{A}_{:j}$.

In terms of the unfolded tensors, (11) and (12) are equivalent. Generally, the dimension of the input space is much larger than the number of the training samples, i.e., $\prod_{n=1}^{N} L_n \gg J$. The fidelity term of (12) is ill-posed, and thus, it is hard to obtain the desired results. Moreover, the predefined arrangement of tensors is not maintained. To reduce the freedom of the model and leverage the structural information in $\mathcal{A}_{:j}$, we assume the weight tensor $\mathcal{W}_{c:}$ to be low rank. Specifically, we assume that $\mathcal{W}_{c:}$ can be decomposed via rank-1 CP decomposition as $\mathcal{W}_{c:} = \mathbf{w}_{c:}^{(1)} \circ \mathbf{w}_{c:}^{(2)} \circ \cdots \circ \mathbf{w}_{c:}^{(N)}$. Then, problem (12) has $C \sum_{n=1}^{N} L_n$ degrees of freedom.

As seen from (11) and (12), it is clear that the models of RR and its tensorial version are a combination of multiple binary classifiers via one-versus-rest strategy. In other words, their training process can be treated as training C classifiers for C classes separately. We consider coupling these separated classifiers by factorizing $\mathcal{W}_{c:} = \mathbf{w}_{c:}^{(1)} \circ \mathbf{w}^{(2)} \circ \cdots \circ \mathbf{w}^{(N)}$, and thereby reduce the degrees of freedom to $CL_1 + \sum_{n=2}^{N} L_n$.

For problem (12), we have the following formulation by substituting the CP decomposition in the fidelity term:

$$\begin{aligned} f(\mathcal{W}_{c:}) &= \sum_{c,j} (t_{cj} - \langle \mathcal{A}_{:j}, \mathcal{W}_{c:} \rangle)^2 \\ &= \sum_{c,j} \left(t_{cj} - \left\langle \mathcal{A}_{:j}, \mathbf{w}_{c:}^{(1)} \circ \mathbf{w}^{(2)} \circ \cdots \circ \mathbf{w}^{(N)} \right\rangle\right)^2 \\ &= \sum_{c,j} \left(t_{cj} - \mathcal{A}_{:j} \times_1 \mathbf{w}_{c:}^{(1)} \prod_{n=2}^{N} \times_n \mathbf{w}^{(n)}\right)^2 \\ &= \|\mathcal{T} - \mathcal{A} \prod_{n=1}^{N} \times_n \mathbf{W}^{(n)}\|_F^2 \end{aligned} \tag{13}$$

where $\mathcal{T} \in \mathbb{R}^{C \times 1 \times \cdots \times 1 \times J}$ is an $(N+1)$th-order tensor that is constructed by adding singleton dimensions to $\mathbf{T} \in \mathbb{R}^{C \times J}$, $\mathcal{A} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N \times J}$ contains the input features of all training samples, the cth row of $\mathbf{W}^{(1)} \in \mathbb{R}^{C \times L_1}$ is equal to $\mathbf{w}_{c:}^{(1)}$, and $\mathbf{W}^{(n)} \in \mathbb{R}^{1 \times L_n}$ is the transpose of $\mathbf{w}^{(n)}$ ($n \in \{2, \ldots, N\}$). Similarly, the regularization term of (12) can be rewritten as

$$\begin{aligned} g(\mathcal{W}_{c:}) &= \sum_c \|\mathcal{W}_{c:}\|_F^2 \\ &= \sum_c \left\|\mathbf{w}_{c:}^{(1)} \circ \mathbf{w}^{(2)} \circ \cdots \circ \mathbf{w}^{(N)}\right\|_F^2 \\ &= \sum_c \sum_{l_1, l_2, \ldots, l_N} \left(w_{cl_1}^{(1)} w_{l_2}^{(2)} \ldots w_{l_N}^{(N)}\right)^2 \\ &= \sum_c \left\|\mathbf{w}_{c:}^{(1)}\right\|_2^2 \cdot \|\mathbf{w}^{(2)}\|_2^2 \ldots \|\mathbf{w}^{(N)}\|_2^2 \\ &= \prod_{n=1}^{N} \|\mathbf{W}^{(n)}\|_F^2. \end{aligned} \tag{14}$$

Then, problem (12) can be rewritten as

$$\min_{\mathbf{W}^{(n)}} \ \left\|\mathcal{T} - \mathcal{A} \prod_{n=1}^{N} \times_n \mathbf{W}^{(n)}\right\|_F^2 + \lambda \prod_{n=1}^{N} \left\|\mathbf{W}^{(n)}\right\|_F^2. \tag{15}$$

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Notably, if the input features are vectors, problem (15), i.e., tensor RR (TRR), can degrade into the RR model (8).

In the above model (15), the weight matrices $\mathbf{W}^{(n)}$ are independent of each other, since the order of the multiple n-mode products is irrelevant in the fidelity term and each $\mathbf{W}^{(n)}$ is decoupled from $\mathcal{W}_{c:}$ in the regularization term. That is to say, we can treat each mode of $\mathcal{A}$ separately and explore prior knowledge for each $\mathbf{W}^{(n)}$. For $\mathbf{W}^{(1)}$, if the input features are vectors, it is equal to the weight matrix $\mathbf{W} \in \mathbb{R}^{C\times L}$ of (8), and thus we keep it unchanged. For $\{\mathbf{W}^{(n)}|n = 2,\ldots,N\}$, they can be treated as the weights of the mode-1 fibers $\mathbf{a}_{:l_2\ldots l_N}$ of $\mathcal{A}_{:j}$, and these mode-1 fibers taken from different modes can be seen as the instances of different spatial directions or different tasks. It is natural to enforce the weight matrices $\{\mathbf{W}^{(n)}|n = 2,\ldots,N\}$ to be nonnegative, since each mode-1 fiber should contribute to the final results. Moreover, the nonnegative constraint can promote the sparsity of the weight matrices [45]–[47], and thus it further reduces the model's degrees of freedom and is beneficial for the optimal selection of the mode-1 fibers. For the tensorial classification, the nonnegativity of weight matrices provides it with the ability to capture the generalized relationship of input tensors, i.e., the optimal mode-1 fibers, on the basis that CP decomposition leverages the structural information. By introducing the nonnegative constraint, problem (15) can be written as

$$\begin{aligned}&\min_{\mathbf{W}^{(n)}} \left\|\mathcal{T} - \mathcal{A}\prod_{n=1}^{N}\times_n\mathbf{W}^{(n)}\right\|_F^2 + \lambda\prod_{n=1}^{N}\left\|\mathbf{W}^{(n)}\right\|_F^2\\ &\text{s.t. } \mathbf{W}^{(n)} \succeq 0, \quad n = 2,\ldots,N\end{aligned} \tag{16}$$

where $\mathbf{W}^{(n)} \succeq 0$ means that all elements of $\mathbf{W}^{(n)}$ are nonnegative.

In [42]–[44], they suggest that rank-1 CP decomposition is too restrictive, which may lead to loss of discriminative information. In this article, we also consider imposing rank-R constraint in (16) to decrease the regression error by factorizing $\mathcal{W}_{c:} = \sum_{r=1}^{R}\mathcal{W}_{r,c:} = \mathbf{w}_{r,c:}^{(1)} \circ \mathbf{w}_r^{(2)} \circ \cdots \circ \mathbf{w}_r^{(N)}$, and then conduct model estimation by solving the following optimization problem:

$$\begin{aligned}&\min_{\mathcal{W}_{r,c:}} \sum_{c,j}\left(t_{cj} - \left\langle\mathcal{A}_{:j}, \sum_r \mathcal{W}_{r,c:}\right\rangle\right)^2 + \sum_{c,r}\lambda_r\|\mathcal{W}_{r,c:}\|_F^2\\ &\text{s.t. } \mathcal{W}_{r,c:} = \mathbf{w}_{r,c:}^{(1)}\prod_{n=2}^{N}\circ\mathbf{w}_r^{(n)}, \quad \mathbf{w}_r^{(n)} \succeq 0,\ n = 2,\ldots N\end{aligned} \tag{17}$$

where $\{\lambda_r > 0|r = 1, 2\ldots, R\}$ are regularization parameters. Note that if $R = 1$, (17) is exactly (16).

For a test sample $\mathbf{x} \in \mathbb{R}^L$ in vector space, we extend it to the tensor space as $\mathcal{X} \in \mathbb{R}^{L_1\times L_2\times\cdots\times L_N}$. Then, its label can be determined by

$$y = \arg\max_{c=1,2,\ldots,C} \delta_c\left(\sum_r \mathcal{X}\prod_{n=1}^{N}\times_n\mathbf{W}_r^{(n)}\right) \tag{18}$$

where $\mathbf{W}_r^{(1)} = [\mathbf{w}_{r,1:}^{(1)},\ldots,\mathbf{w}_{r,C:}^{(1)}]^T$ and $\mathbf{W}_r^{(n)} = (\mathbf{w}_r^{(n)})^T$ $(n \in \{2,\ldots,N\})$.

Algorithm 1 Rank-1 GTR

1: **Input**: Test tensor: $\mathcal{X} \in \mathbb{R}^{L_1\times L_2\times\cdots\times L_N}$; training tensors: $\mathcal{A} \in \mathbb{R}^{L_1\times L_2\times\cdots\times L_N\times J}$; label tensor: $\mathcal{T} \in \mathbb{R}^{C\times 1\times\cdots\times 1\times J}$; parameter: λ.
2: % Initialization
3: Construct $\mathbf{T}$ by removing the singleton dimensions of $\mathcal{T}$.
4: **for** $n = 2 : N$ **do**
5: Compute the mean of $\mathcal{A}_{:j}$ on all modes but the nth
6: Initialize $\mathbf{W}^{(n)}$ by (9)
7: **end for**
8: % Main algorithm
9: **while** The stopping conditions are not satisfied **do**
10: Compute ω_1 and $\mathcal{B}^{(1)}$
11: Update $\mathbf{W}^{(1)}$ by (21)
12: **for** $n = 2 : N$ **do**
13: Compute ω_n and $\mathcal{B}^{(n)}$
14: Update $\mathbf{W}^{(n)}$ by (22)
15: **end for**
16: **end while**
17: **Output**: The estimated label y of $\mathcal{X}$ according to (18).

B. Optimization Algorithm for GTR

For problem (16), it is hard to obtain the weight matrices $\mathbf{W}^{(n)}$ directly, since they are coupled with each other. In this article, we consider using alternating update to solve (16), which updates one variable by fixing other variables. By recalling (6), we can rewrite the fidelity term as

$$\begin{aligned}f(\mathcal{W}_{c:}) &= \|\mathcal{T} - \mathcal{B}^{(n)}\times_n\mathbf{W}^{(n)}\|_F^2\\ &= \left\|\mathbf{T}_{(n)} - \mathbf{W}^{(n)}\mathbf{B}_{(n)}^{(n)}\right\|_F^2\end{aligned} \tag{19}$$

where $\mathcal{B}^{(n)} = \mathcal{A}\prod_{i=1,i\neq n}^{N}\times_i\mathbf{W}^{(i)}$. Then, the nth subproblem of (16) without the nonnegative constraint can be rewritten as

$$\min_{\mathbf{W}^{(n)}} \left\|\mathbf{T}_{(n)} - \mathbf{W}^{(n)}\mathbf{B}_{(n)}^{(n)}\right\|_F^2 + \lambda\omega_n\|\mathbf{W}^{(n)}\|_F^2 \tag{20}$$

where $\omega_n = \prod_{i=1,i\neq n}^{N}\|\mathbf{W}^{(i)}\|_F^2$. With $\{\mathbf{W}^{(i)}|i \neq n, i = 1,\ldots,N\}$ fixed, the solution of problem (20) can be written as

$$\mathbf{W}^{(n)} = \mathbf{T}_{(n)}\left(\mathbf{B}_{(n)}^{(n)}\right)^T\left[\mathbf{B}_{(n)}^{(n)}\left(\mathbf{B}_{(n)}^{(n)}\right)^T + \lambda\omega_n\mathbf{I}\right]^{-1}. \tag{21}$$

If the nonnegative constraint is imposed in (20), the solution can be simply written as

$$\mathbf{W}^{(n)} = \max\left\{\mathbf{T}_{(n)}\left(\mathbf{B}_{(n)}^{(n)}\right)^T\left[\mathbf{B}_{(n)}^{(n)}\left(\mathbf{B}_{(n)}^{(n)}\right)^T + \lambda\omega_n\mathbf{I}\right]^{-1}, 0\right\} \tag{22}$$

where $\max\{\cdot, 0\}$ is used to set the negative components to zero and keep the nonnegative components unchanged. The complete algorithm for solving problem (16) is summarized in Algorithm 1. For a warm start, we initialize $\mathbf{W}^{(n)}$ by using RR, as shown in steps 3–7 of Algorithm 1.

Compared with (16), problem (17) is more difficult to solve because: 1) the rank R of CP decomposition needs to be prespecified; 2) there are many variables $\{\mathbf{W}_r^{(n)}|r = 1,\ldots,R; n = 1,\ldots,N\}$ to be solved simultaneously; and 3) the model involves many regularization parameters $\{\lambda_r|r = 1, 2\ldots, R\}$, for which the joint tuning

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Algorithm 2 Rank-R GTR

Input: Test tensor: $\mathcal{X} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N}$; training tensors: $\mathcal{A} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_N \times J}$; label tensor: $\mathcal{T} \in \mathbb{R}^{C \times 1 \times \cdots \times 1 \times J}$; parameters: R and $\{\lambda_r | r = 1, \ldots, R\}$.
Set $\mathcal{T}_1 = \mathcal{T}$
for $r = 1 : R$ **do**
  Solve the subproblem (23) by Algorithm 1
  $\mathcal{T}_{r+1} = \mathcal{T}_r - \mathcal{A} \prod_{n=1}^{N} \times_n \mathbf{W}_r^{(n)}$
end for
Output: The estimated label y of $\mathcal{X}$ according to (18).

becomes time consuming. In this article, we consider using a divide-and-conquer strategy [44], [48] to solving (17), which seeks a rank-1 tensor at a time and then deflates to find further rank-1 tensors from the residuals. Specifically, we sequentially solve the following optimization problem:

$$\begin{aligned} &\min_{\mathcal{W}_{r,c:}} \sum_{c,j} (t_{r,cj} - \langle \mathcal{A}_{:j}, \mathcal{W}_{r,c:} \rangle)^2 + \lambda_r \|\mathcal{W}_{r,c:}\|_F^2 \\ &\text{s.t. } \mathcal{W}_{r,c:} = \mathbf{w}_{r,c:}^{(1)} \prod_{n=2}^{N} \circ \mathbf{w}_r^{(n)}, \quad \mathbf{w}_r^{(n)} \succeq 0, \ n = 2, \ldots N \end{aligned} \tag{23}$$

where r denotes the sequential number and $t_{r,cj}$ is the current residue of label with

$$t_{r,cj} = \begin{cases} t_{cj}, & \text{if } r = 1 \\ t_{r-1,cj} - \langle \mathcal{A}_{:j}, \mathcal{W}_{r-1,c:} \rangle, & \text{otherwise} \end{cases} \quad \forall c, j \tag{24}$$

where $\mathcal{W}_{r-1,c:}$ is the estimated rank-1 tensor in the last step. The subproblem (23) of (17) is equivalent to problem (16), and thus, it can be solved by Algorithm 1. The complete algorithm for solving problem (17) is summarized in Algorithm 2.

C. Instantiations of GTR

The tensor learning methods usually seek the tensorial features in a physical sense [27], [28]. For example, the tensorial features are ubiquitous, and any set of features that follows a predefined arrangement can be treated as a meaningful tensor. To evaluate the ability of GTR to deal with generalized tensors, we construct tensorial features from different views and thereby investigate six classical instantiations. The details of these features are described as follows.

1) The first one is the spatially derived (SD) features. This kind of tensorial features exists physically and is also the major one of the existing tensorial classification methods [27], [28]. Specifically, we arrange each pixel and its surrounding pixels as a third-order tensor, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{L \times L_w \times L_h}$ with L_w and L_h being the sizes of a spatial window.
2) The second one is the multiple filtering (MF) features. These features are extracted by sequentially performing the mean filtering on each band of the original hyperspectral image, and the tensors are constructed by varying the window size of the mean filtering, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{L \times L_f}$ with L_f being the number of mean filters.
3) The third one is the multiple scale (MS) features. First, a superpixel-level multiscale segmentation is performed on the original hyperspectral images to obtain several nonoverlapping spectrally similar regions for each scale [21]. Then, the mean filtering is performed within the regions of each scale to generate the corresponding features. Similar to the MF features, the tensors are constructed by varying the scale size of the segmentation, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{L \times L_s}$ with L_s being the number of scales.
4) The fourth one is the multiple-kernel (MK) features. Since the kernel tensor analysis is beyond the scope of this article, one simple way to make use of the available kernel matrices is to directly take the columns of each kernel matrix as features, i.e., the column generation (CG) strategy. Specifically, the tensors are constructed by varying the parameters of the kernel function, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{J \times L_k}$ with L_k being the number of kernel parameters.
5) The fifth one is the MF kernel (MFK) features. This kind of tensorial features is similar to the MF features with each mode-1 fiber replaced by kernel vector, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{J \times L_f}$.
6) The last one is the MS kernel (MSK) features. Similar to the MFK features, this kind of tensorial features is constructed by replacing mode-1 fiber of the MS features with kernel vector, i.e., $\mathcal{A}_{:j}, \mathcal{X} \in \mathbb{R}^{J \times L_s}$.

IV. Experimental Results and Analysis

In this section, we first introduce three hyperspectral imagery data sets used in experiments and then evaluate the proposed GTR approach by a series of experiments. Three widely used quality metrics are utilized to evaluate the performance of the classification methods, i.e., the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (KC). Before our experiments, the original data have been scaled in the range $[0, 1]$ and, unless otherwise specified, the quantitative measures are obtained by averaging ten random sampling runs. In addition, all experiments are carried out in a 64-b quad-core CPU 3.40-GHz processor with 12-GB memory.

A. Hyperspectral Imagery Data Sets

To evaluate the performance of the proposed approach, three hyperspectral imagery data sets have been considered in our experiments.

1) The first data set is the Indian Pines (IndianP) image acquired by the airborne visible/infrared imaging spectrometer (AVIRIS). This image, with the size of 145×145 pixels, contains 220 bands covering the wavelength range of 0.4–2.5 μm, and the spectral and spatial resolutions are 10 nm and 17 m, respectively. There are 16 ground reference classes of interests, as shown in Table I. The false-color image and the ground reference map are shown in Fig. 1. In our experiments, there are 200 bands retained by removing 20 water absorption bands. For each class, we randomly

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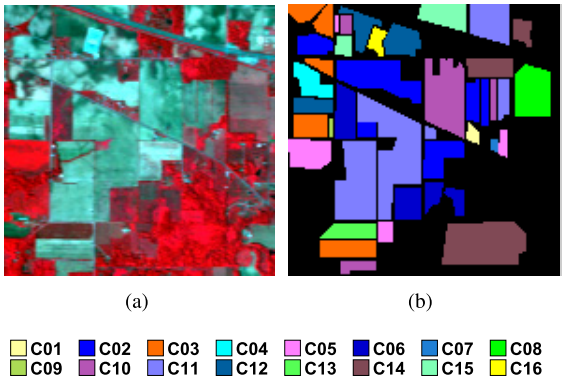


Fig. 1. AVIRIS IndianP data set. (a) RGB composite image of three bands. (b) Ground reference map.

TABLE I

SIXTEEN GROUND REFERENCE CLASSES IN THE AVIRIS INDIANP DATA SET

NO	Name	Samples
C01	Alfalfa	54
C02	Corn-no till	1434
C03	Corn-min till	834
C04	Corn	234
C05	Grass/pasture	497
C06	Grass/trees	747
C07	Grass/pasture-mowed	26
C08	Hay-windrowed	489
C09	Oats	20
C10	Soybeans-no till	968
C11	Soybeans-min till	2468
C12	Soybean-clean till	614
C13	Wheat	212
C14	Woods	1294
C15	Bldg-grass-tree drives	380
C16	Stone-steel towers	95
Total		10366

TABLE II

NINE GROUND REFERENCE CLASSES IN THE ROSIS PAVIAU DATA SET

NO	Name	Samples
C1	Asphalt	6852
C2	Meadow	18686
C3	Gravel	2207
C4	Trees	3436
C5	Metal sheets	1378
C6	Bare soil	5104
C7	Bitumen	1356
C8	Bricks	3878
C9	Shadows	1026
Total		43923

choose 1%, 3%, 5%, 10%, 15%, and 20% of the labeled pixels for training and the remaining pixels for testing. As for very small classes, we take a minimum of two training pixels per class.

2) The second data set is the University of Pavia (PaviaU) image acquired by the reflective optics system imaging spectrometer (ROSIS). The ROSIS sensor collects 115 bands, covering the wavelength range from 0.43 to 0.86 μm. This image contains 610×340 pixels, with a very high spatial resolution of 1.3 m per pixel. There are nine ground reference classes of interests, as shown in Table II. The false-color image and the ground reference map are shown in Fig. 2. In our experiments, we work with 103 spectral bands after the removal of noisy bands and build training sets by randomly choosing 10, 20, 40, 60, 80, and 100 training pixels per class.

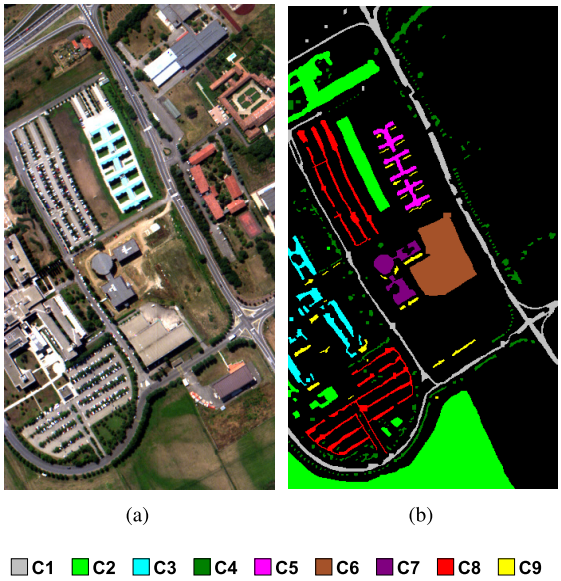


Fig. 2. ROSIS PaviaU data set. (a) RGB composite image of three bands. (b) Ground reference map.

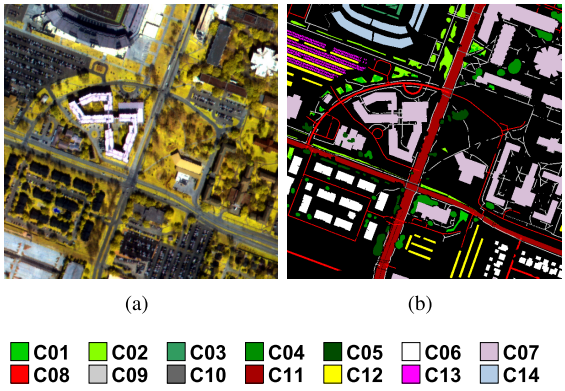


Fig. 3. ITRES CASI-1500 UH data set. (a) RGB composite image of three bands. (b) Ground reference map.

3) The third data set is the University of Houston (UH) image acquired by an ITRES CASI-1500 sensor, covering a 0.38–1.05-μm spectral range with 48 bands at a 1-m ground sampling distance. This image is one of the multimodal optical remote sensing data sets released by the 2018 Data Fusion Contest of the IEEE Geoscience and Remote Sensing Society (GRSS) [49]. These data sets available on the website[1] were acquired by the National Center for Airborne Laser Mapping (NCALM) at UH on February 16, 2017, covering the UH campus and its surrounding urban areas. For the considered hyperspectral imagery, its original image size is 4172×1202 pixels, and a subimage with a size of 601×601 is selected. This subimage contains 14 ground reference classes of interests, as shown in Table III. The false-color image and the ground reference map are

[1] http://hyperspectral.ee.uh.edu/?page_id=1075

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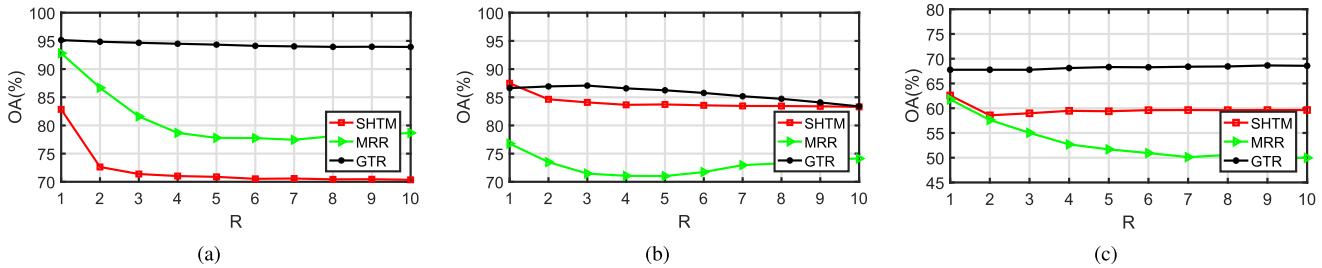


Fig. 4. OA as a function of R when applied to the (a) AVIRIS IndianP data set, (b) ROSIS PaviaU data set, and (c) ITRES CASI-1500 UH data set.

TABLE III
FOURTEEN GROUND REFERENCE CLASSES IN THE ITRES CASI-1500 UH DATA SET

NO	Name	Samples
C01	Healthy grass	1649
C02	Stressed grass	5816
C03	Artificial turf	684
C04	Evergreen trees	4903
C05	Deciduous trees	3478
C06	Residential buildings	11441
C07	Non-residential buildings	41163
C08	Roads	7549
C09	Sidewalks	11239
C10	Crosswalks	371
C11	Major thoroughfares	17877
C12	Paved parking lots	5289
C13	Cars	3372
C14	Stadium seats	6824
Total		121655

shown in Fig. 3. In our experiments, we build training sets by randomly choosing 10, 20, 40, 60, 80, and 100 training pixels per class and use the rest as test sets.

B. Parameter Setting for Different Instantiations

In this section, we will introduce the parameters used in our experiments for the six instantiations described in Section III-C. As for other parameters brought by the classifiers, they are chosen by cross validation. Since we do not focus on extracting features, most parameters of the six instantiations are chosen empirically. For the SD features, the spatial sizes of a window are fixed to 9×9 for all data sets, i.e., $L_w = L_h = 9$. For the MF features used in all data sets, the multiple mean filtering is achieved by varying the radius of a square window from 0 to 10 in steps of 1, i.e., $L_f = 11$. For the MS features, the free parameters are fixed to be the same as those used in [21]. Including the original image, there are totally 31 scales for all data sets, i.e., $L_s = 31$. For the MK features, only the Gaussian radial basis function (RBF) kernel ($K(\mathbf{x}, \mathbf{y}) = \exp(-\gamma\|\mathbf{x} - \mathbf{y}\|_2^2), \gamma > 0$) is considered, and the RBF-kernel parameter is varied in the range $\gamma \in \{2^{-5}, 2^{-4.5}, \ldots, 2^5\}$, i.e., $L_k = 21$. As for the MFK and MSK features, the parameters of multiple representation are as same as those of the MF and MS features, respectively. The RBF-kernel is adopted in these two kinds of tensorial features, and γ is chosen by cross validation.

C. Influence of Rank R

In traditional tensorial methods, it is suggested to perform CP decomposition using higher ranks to capture more discriminative information. This set of experiments is used to show the influence of rank R for hyperspectral image classification. Besides the proposed GTR, another two tensor-based classifiers are investigated, i.e., support higher order tensor machine (SHTM) [41] and multiple rank regression (MRR) [43]. For SHTM, the CP decomposition is implemented by the MATLAB Tensor Toolbox [50], and the main algorithm is achieved by the LIBSVM software package [51]. For MRR, the code is provided in [43]. The adopted tensorial features are the SD features, since they are widely used in remote sensing community. For the number of training samples, 5% of the labeled pixels are used for the AVIRIS IndianP data set, and 40 for the ROSIS PaviaU data set and the ITRES CASI-1500 UH data set. Fig. 4 shows the impact of R for the three given data sets, where R is varied from 1 to 10. It can be seen that in most cases, rank-1 decomposition performs better than the others for the three data sets and all methods. For exceptional cases, the improvements are negligible. Moreover, Fig. 5 shows the regression error (i.e., the value of fidelity term) of the proposed GTR as a function of the number of iterations for all ranks R, where for each rank-1 decomposition, the number of iterations is fixed to 100. It is clear that in each rank-1 decomposition, the algorithm converges very quickly and the regression error is reduced. In other words, although rank-1 decomposition is too restrictive, it benefits the classification purpose. For this, rank-1 decomposition is used for all methods in the following experiments, i.e., $R = 1$.

D. Influence of Nonnegativity and Initialization

This set of experiments is designed to analyze the impact of the nonnegative constraint by random initialization. Specifically, we compare the proposed GTR method with its degenerated version TRR on the six instantiations. Thus, this set of experiments can also be utilized to evaluate the ability of GTR to deal with generalized tensors. For both methods, λ is set to be equal to each other, and for each random sampling, the weight matrices $\mathbf{W}^{(n)}$ are randomly initialized 50 times to observe the numerical and statistical differences. For the number of training samples, 1% and 5% of the labeled pixels are used for the AVIRIS IndianP data set, and 10 and 40 for the ROSIS PaviaU data set and the ITRES CASI-1500 UH data set. Fig. 6 shows the results of two compared methods, where "-1" denotes the methods using 1% or 10 training samples and "-3" denotes the methods

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TABLE IV

CLASSIFICATION ACCURACIES FOR THE AVIRIS INDIANP DATA SET USING DIFFERENT CLASSIFICATION METHODS

Class	SD				MF				MS				MK				MFK				MSK			
	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR
C01	38.82	24.71	77.06	53.73	67.45	50.78	81.76	73.73	91.18	**97.84**	94.51	92.94	49.02	50.59	50.00	53.53	53.33	56.27	68.82	82.75	91.37	96.08	92.16	92.35
C02	82.08	93.45	93.16	94.68	90.79	94.57	94.14	94.62	93.18	95.40	96.31	**96.48**	64.84	67.96	67.78	72.79	72.14	78.36	85.63	90.95	93.36	94.44	95.45	96.48
C03	82.34	84.89	90.45	90.03	92.17	95.54	91.41	93.99	98.61	**100**	99.80	**100**	65.18	60.20	59.36	64.76	74.14	66.38	87.03	93.69	98.36	99.38	99.28	99.92
C04	38.47	83.02	85.23	88.11	67.12	**92.16**	84.82	91.71	84.05	89.95	92.07	90.00	38.51	36.22	36.85	44.32	53.24	50.27	63.15	78.51	90.54	88.56	91.85	91.49
C05	84.98	89.58	91.55	93.18	91.14	92.92	93.98	93.94	95.34	95.53	95.47	**97.18**	83.41	84.28	84.68	87.08	80.59	87.61	91.23	92.58	95.44	94.89	95.81	96.14
C06	89.62	99.35	98.56	99.31	96.04	99.42	**99.44**	99.37	98.53	99.01	98.94	98.21	84.13	90.86	91.62	93.74	86.40	96.15	98.00	98.93	98.96	98.79	98.79	98.87
C07	79.58	10.00	85.00	88.33	96.25	45.00	89.17	83.33	96.67	96.67	96.67	96.67	74.17	85.83	80.00	85.42	83.75	51.67	87.92	**98.75**	96.67	96.67	96.67	96.67
C08	95.45	99.87	99.74	99.96	96.49	**100**	99.96	99.98	99.57	99.81	99.76	99.78	89.68	95.56	96.25	97.05	93.19	98.69	97.54	99.27	99.78	99.96	99.78	99.78
C09	44.44	14.44	59.44	46.11	64.44	16.67	47.78	36.67	87.22	93.33	93.33	90.00	27.78	39.44	33.89	45.56	46.67	63.33	81.67	89.44	**100**	**100**	96.67	**100**
C10	77.15	90.76	91.16	91.73	86.01	90.58	89.70	90.76	90.55	90.73	95.87	94.42	63.75	71.55	67.95	73.24	65.68	72.39	82.79	88.54	95.52	95.28	95.63	**96.26**
C11	86.58	94.98	94.74	96.25	93.88	97.57	96.13	97.38	91.41	99.09	98.63	**99.35**	73.45	78.52	75.55	80.06	74.50	82.03	88.09	93.95	95.56	98.34	98.31	98.58
C12	71.89	92.26	92.90	96.30	93.10	96.66	94.60	96.48	91.96	**97.77**	97.75	96.76	57.48	53.67	55.85	63.74	57.48	80.57	79.98	88.25	93.09	95.64	96.60	96.28
C13	87.11	97.36	97.16	98.01	96.87	98.61	98.81	98.26	99.05	99.50	99.25	**99.60**	87.41	94.83	95.12	96.47	91.79	98.86	98.21	99.30	99.50	99.30	99.50	99.40
C14	92.42	99.32	98.71	99.52	97.55	99.75	99.36	99.72	99.85	99.10	99.96	99.94	89.11	93.81	93.70	94.34	90.63	95.50	96.62	97.95	99.93	99.28	**100**	**100**
C15	62.91	83.60	91.16	92.02	85.84	93.88	92.19	96.18	92.80	96.81	95.07	95.60	40.94	36.48	39.03	45.96	66.29	62.13	78.53	83.60	96.48	94.18	94.96	**96.84**
C16	85.56	88.44	93.56	97.78	96.89	**99.44**	94.33	98.67	96.11	96.44	96.67	97.78	86.78	49.89	89.67	87.44	87.00	58.33	70.44	75.67	95.22	90.33	97.44	96.33
OA(%)	82.86	92.81	94.10	95.15	92.14	95.84	94.96	96.04	94.41	97.32	97.86	97.88	71.93	74.71	74.25	78.28	75.91	81.47	88.08	92.87	96.30	97.06	97.54	**97.92**
AA(%)	74.96	77.88	89.97	89.06	88.25	85.22	90.47	90.30	94.13	96.69	96.88	96.54	67.23	68.11	69.83	74.09	73.55	74.91	84.73	90.76	96.24	96.32	96.81	**97.21**
KA(%)	80.49	91.79	93.29	94.47	91.05	95.26	94.25	95.48	93.64	96.95	97.56	97.59	68.10	71.07	70.96	75.20	72.64	78.79	86.41	91.87	95.79	96.65	97.20	**97.63**
Time(s)	203	41	218	7	145	7	169	**1**	143	19	177	14	296	43	847	31	256	24	931	18	288	50	879	38

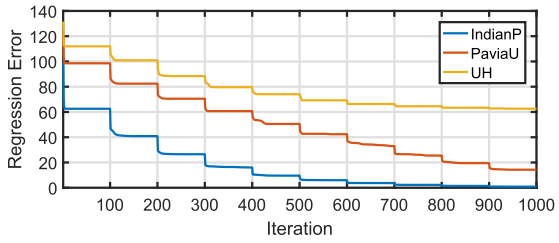


Fig. 5. Regression error as a function of the number of iterations when applied to the three given data sets.

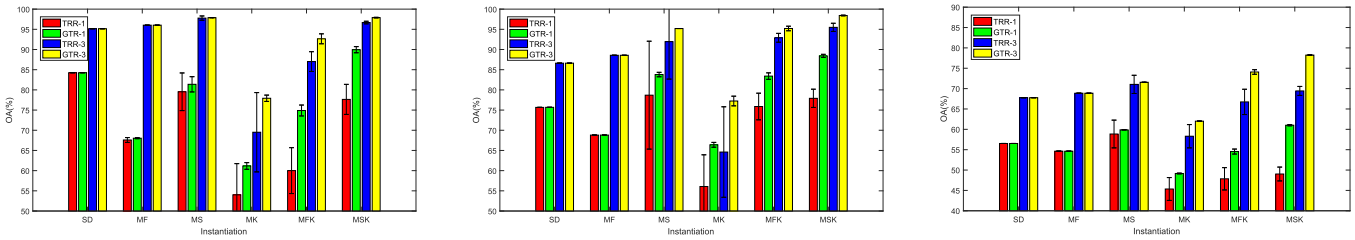
using 5% or 40 training samples. In this figure, the error bars indicate the standard deviation for 50 random runs. It is clear that for the first two instantiations, the accuracies of GTR and TRR are almost the same, and for the remaining four instantiations, GTR performs better than TRR in terms of both the accuracy and standard deviation. The standard deviations of GTR are very small in almost all cases, while the standard deviations of TRR are large for the last four instantiations, especially when the number of training samples is small. In other words, GTR is not sensitive to the initial values, and TRR is not a stable method. Moreover, for the last three kinds of generalized tensors, the improved performance of GTR over TRR indicates that GTR achieves the optimal selection of the mode-1 fibers.

E. Comparisons With Respect to Different Instantiations

Here, we evaluate the proposed approach by comparing GTR with several related classification methods on the six instantiations described in Section III-C. According to the six instantiations, the compared methods can be divided into six categories.

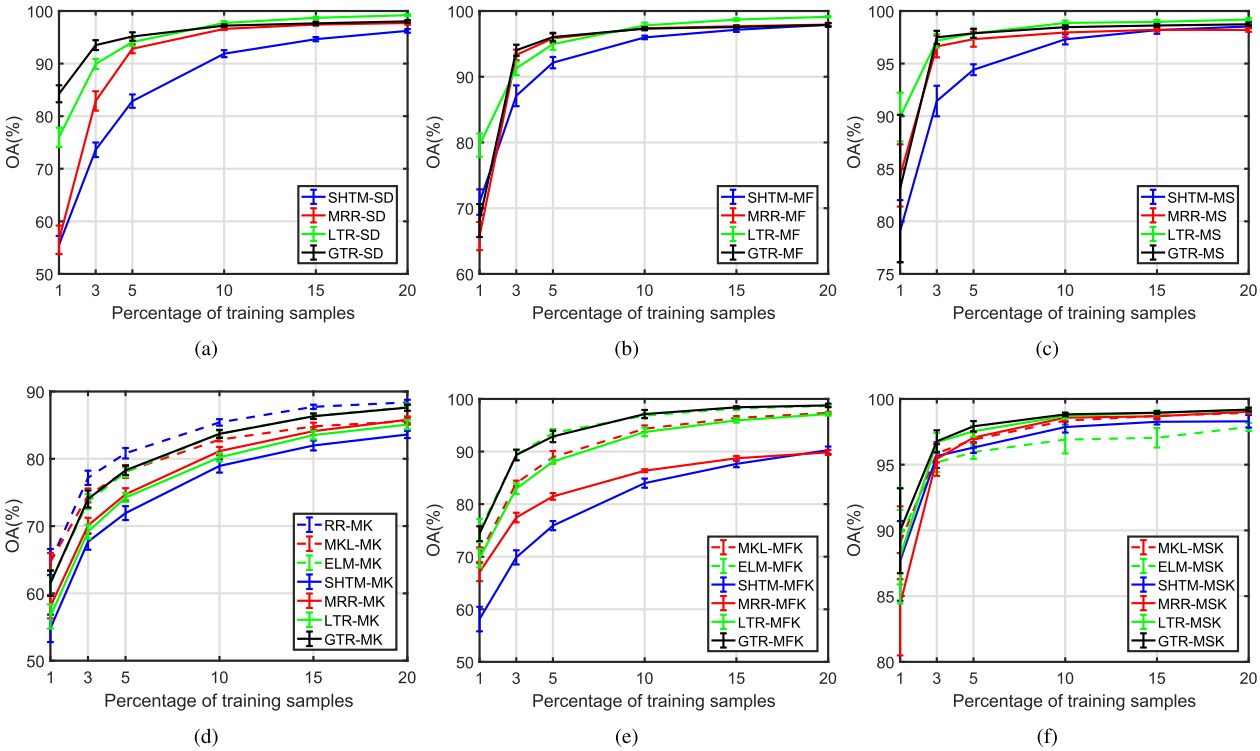
1) The first category comprises four tensor-based classifiers based on the SD features, including SHTM, MRR, LTR, and the proposed GTR. The implementation details of the SHTM and MRR methods used in this section are described in Section IV-C. As for LTR, it is implemented according to [28], and the logistic regression problem is solved by using bound optimization algorithms [52]. For the sake of simplicity, these methods are abbreviated by SHTM-SD, MRR-SD, LTR-SD, and GTR-SD.
2) The second category comprises four methods based on the MF features. They are the four methods mentioned in the first category, and they are abbreviated by SHTM-MF, MRR-MF, LTR-MF, and GTR-MF.
3) Similar to the first two categories, the third category consists of four methods based on the MS features, which are abbreviated by SHTM-MS, MRR-MS, LTR-MS, and GTR-MS.
4) The fourth category comprises seven methods based on the MK features. Besides the aforementioned four tensor-based classifiers, there are two multiple kernel-based methods. One is the class-specific sparse multiple kernel learning method proposed in [53] and [54] denoted by MKL, and the other is the multiple kernel extreme learning machine [55] denoted by ELM. For MKL, the core code is implemented by modifying the LIBSVM software package. For ELM, it is implemented according to [55]. Notably, MKL and ELM directly perform the selection of kernels. That is to say, they are not CG-based methods. In addition, the CG-based RR method using the optimal kernel matrix has been included as a baseline classifier. To distinguish different categories, we add a suffix "−MK" to all methods of this category.
5) The fifth category comprises six methods based on the MFK features. They are the aforementioned methods in the fourth category, i.e., MKL, ELM, SHTM, MRR, LTR, and the proposed GTR method. All these methods have a suffix "−MFK" to distinguish them from the others.
6) Similar to the fifth category, the last category consists of six methods based on the MSK features, which have a suffix "−MSK."

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(a) (b) (c)

Fig. 6. Influence of nonnegativity and initialization. (a) AVIRIS IndianP data set. (b) ROSIS PaviaU data set. (c) ITRES CASI-1500 UH data set.



(a) (b) (c)

(d) (e) (f)

Fig. 7. OA as a function of the number of training samples for six tensorial scenes when applied to the AVIRIS IndianP data set. (a) SD. (b) MF. (c) MS. (d) MK. (e) MFK. (f) MSK.

Fig. 7 illustrates the results of these methods mentioned above as a function of the number of training samples for the AVIRIS IndianP data set. It can be seen that among the tensorial methods, i.e., SHTM, MRR, LTR, and the proposed GTR, GTR outperforms SHTM and MRR for the first three tensorial features and performs best for the last three kinds of tensorial features. The performance of SHTM is lower than the other tensorial methods for all kinds of tensorial features. For the MK and MFK features, GTR performs as well as ELM, and better than MKL. For the MSK features, GTR performs best among the three multiple kernel-based methods. Notably, in this set of experiments and the following, the good performance of GTR on all instantiations over the other tensorial methods indicates the ability of GTR to deal with generalized tensors, and the competitive results of GTR over MKL and ELM further demonstrate that GTR performs a good selection of the mode-1 fibers. Moreover, for the four tensorial methods, Table IV summarizes their global and class-specific accuracies of the usage of 5% training samples, and the processing time in seconds is also included for reference. It is clear that GTR performs best for all kinds of tensorial features, and GTR with the MSK features reports the highest global accuracy. For all cases, GTR needs the lowest computation cost followed by MRR, and LTR is the most time-consuming one. Fig. 8 shows the classification maps obtained for the AVIRIS IndianP data set. These classification maps correspond to one of the ten random runs in order to illustrate the classification scores reported in Table IV. The numerical comparisons shown in Table IV are confirmed by inspecting these classification maps.

Fig. 9 illustrates the results of the aforementioned six categories of methods as a function of the number of training samples for the ROSIS PaviaU data set. It can be seen that for the SD, MF, and MS features, the proposed GTR method provides competitive results, and for the MK, MFK, and MSK features, it outperforms the other three tensorial methods. When compared with the multiple kernel-based methods, the results of GTR are competitive. For the four

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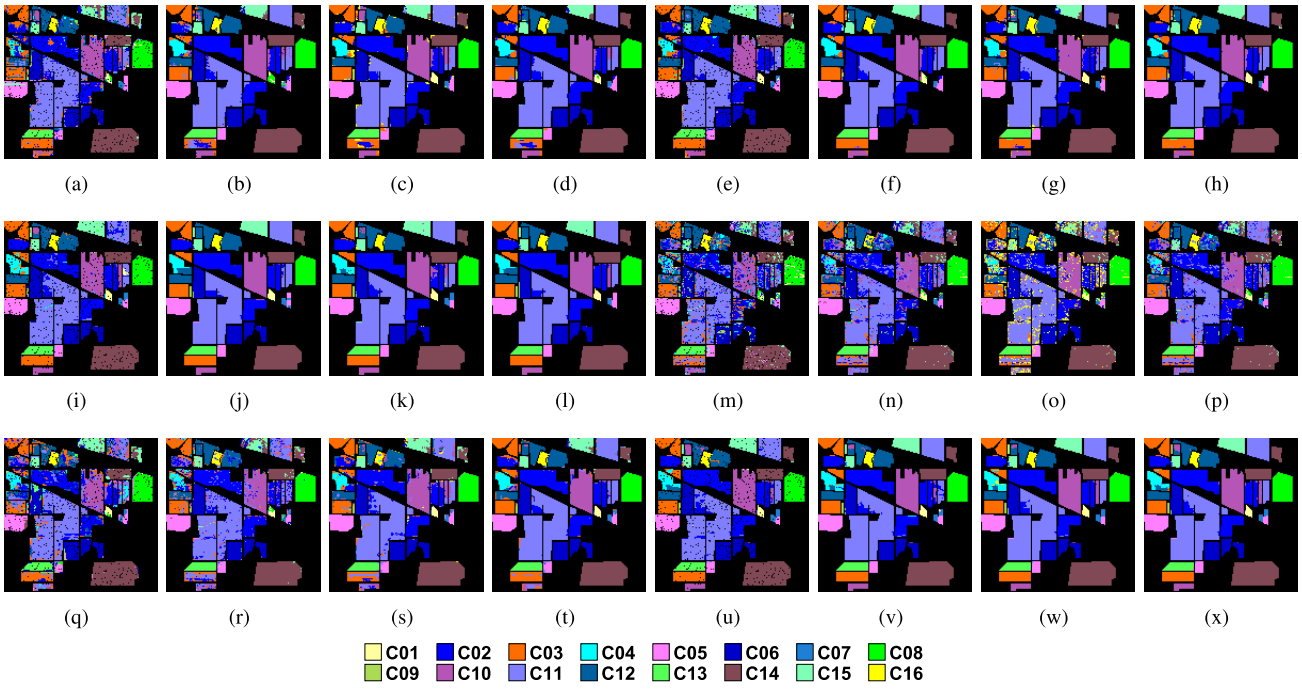


Fig. 8. AVIRIS IndianP data set. (a) SHTM-SD. (b) MRR-SD. (c) LTR-SD. (d) GTR-SD. (e) SHTM-MF. (f) MRR-MF. (g) LTR-MF. (h) GTR-MF. (i) SHTM-MS. (j) MRR-MS. (k) LTR-MS. (l) GTR-MS. (m) SHTM-MK. (n) MRR-MK. (o) LTR-MK. (p) GTR-MK. (q) SHTM-MFK. (r) MRR-MFK. (s) LTR-MFK. (t) GTR-MFK. (u) SHTM-MSK. (v) MRR-MSK. (w) LTR-MSK. (x) GTR-MSK.

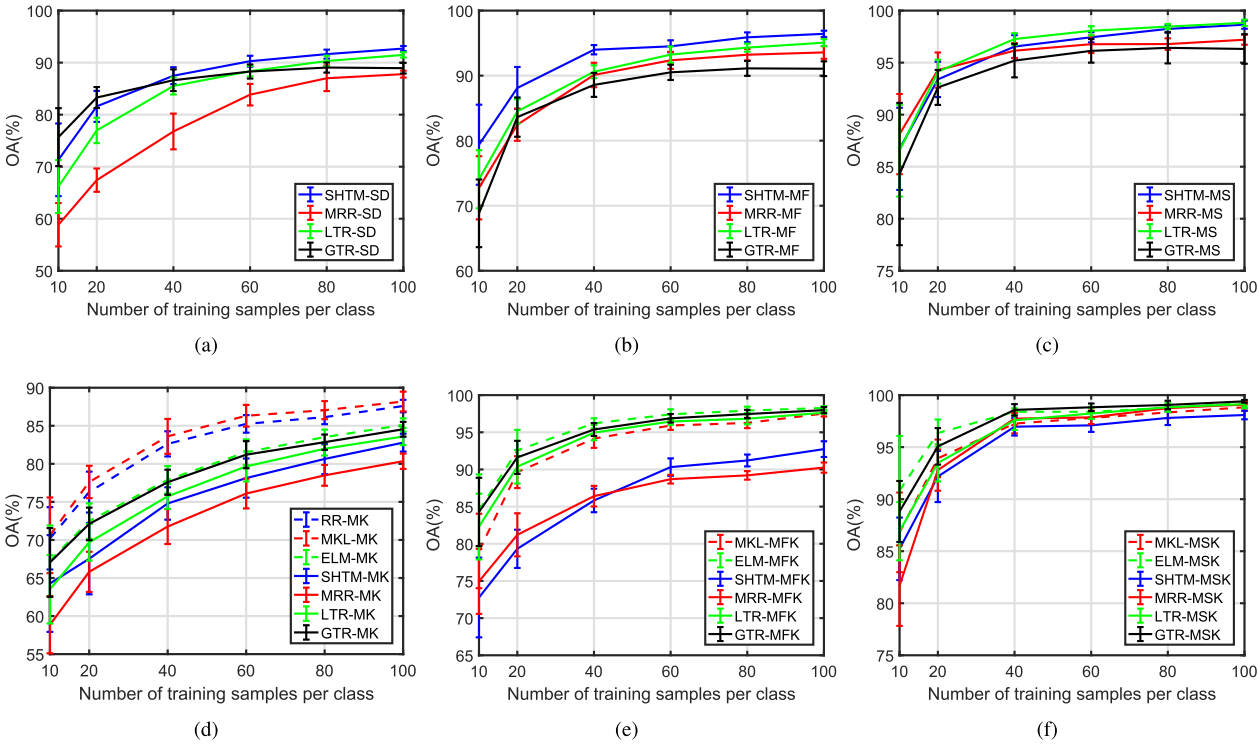


Fig. 9. OA as a function of the number of training samples for six tensorial scenes when applied to the ROSIS PaviaU data set. (a) SD. (b) MF. (c) MS. (d) MK. (e) MFK. (f) MSK.

tensorial methods, Table V reports the global and class-specific accuracies of them, where 40 training samples per class are used. Here, the processing time in seconds is also included for reference. It can be observed that for the first three kinds of tensorial features, the performance of GTR is moderate, while for the remaining kinds of features, GTR is the best. By using the MSK features, GTR provides the highest global accuracy. For the computation cost, it is clear that GTR is the best one and SHTM is the worst one. Fig. 10 shows the classification maps that correspond to one of the ten random runs in each case. These classification maps confirm the numerical results reported in Table V and the validity of the proposed approach.

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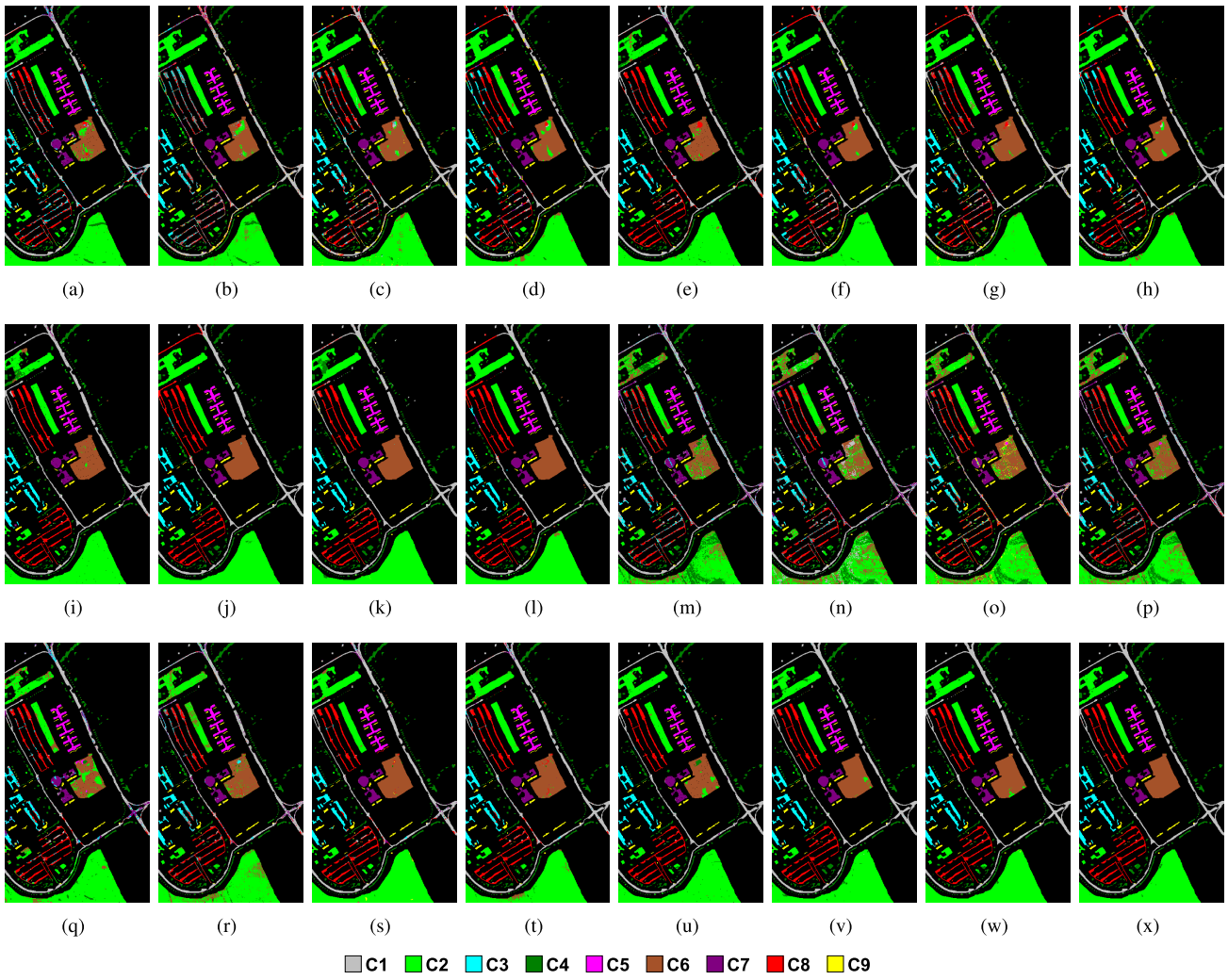


Fig. 10. ROSIS PaviaU data set. (a) SHTM-SD. (b) MRR-SD. (c) LTR-SD. (d) GTR-SD. (e) SHTM-MF. (f) MRR-MF. (g) LTR-MF. (h) GTR-MF. (i) SHTM-MS. (j) MRR-MS. (k) LTR-MS. (l) GTR-MS. (m) SHTM-MK. (n) MRR-MK. (o) LTR-MK. (p) GTR-MK. (q) SHTM-MFK. (r) MRR-MFK. (s) LTR-MFK. (t) GTR-MFK. (u) SHTM-MSK. (v) MRR-MSK. (w) LTR-MSK. (x) GTR-MSK.

TABLE V

CLASSIFICATION ACCURACIES FOR THE ROSIS PAVIAU DATA SET USING DIFFERENT CLASSIFICATION METHODS

Class	SD				MF				MS				MK				MFK				MSK			
	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR
C1	80.92	40.23	70.85	69.37	90.13	77.17	76.13	69.51	97.43	84.68	95.47	90.15	75.52	72.07	70.84	73.59	81.88	77.18	88.25	89.38	97.94	97.55	97.99	**99.04**
C2	92.59	88.07	90.70	92.17	95.75	95.72	96.06	93.93	95.59	98.48	97.98	95.83	69.49	65.44	71.45	72.88	86.55	87.87	97.05	96.76	96.85	97.70	96.93	**98.64**
C3	83.36	85.11	87.43	89.82	90.28	92.65	90.00	91.46	98.18	**99.39**	99.28	99.31	72.10	72.33	73.58	75.72	81.98	82.44	91.85	91.49	97.77	98.97	98.87	98.98
C4	88.21	90.38	86.76	91.74	93.90	96.35	92.72	93.21	93.70	92.22	90.35	86.83	91.51	91.63	91.83	93.11	94.55	92.73	96.03	**96.67**	92.44	93.77	94.51	93.73
C5	99.66	99.51	98.93	**100**	99.90	99.88	99.60	**100**	99.49	99.45	98.98	99.31	98.98	86.13	98.81	99.08	99.72	98.30	99.07	98.74	98.83	99.34	99.25	99.36
C6	84.53	86.21	91.89	91.54	94.45	97.36	97.51	95.96	97.64	99.99	99.99	**100**	69.34	63.36	73.11	76.53	79.79	87.87	97.02	97.49	96.89	98.29	98.79	99.89
C7	86.14	98.49	97.41	99.86	97.08	**99.99**	98.59	99.98	98.81	99.79	99.82	99.73	86.06	89.83	89.73	90.90	86.69	95.36	97.86	99.23	99.10	99.86	99.38	99.79
C8	74.61	36.95	64.87	65.44	89.52	59.97	71.97	71.21	97.75	99.16	97.74	97.16	74.70	77.27	76.74	78.71	83.29	79.94	92.12	94.16	98.34	**99.34**	99.04	99.11
C9	93.42	97.08	93.77	97.56	98.41	**99.66**	95.17	95.74	97.38	97.03	94.08	93.86	98.83	97.52	99.21	99.03	98.68	98.85	95.62	98.63	95.81	96.77	98.01	98.53
OA(%)	87.48	76.77	85.51	86.62	93.98	90.10	90.59	88.58	96.55	96.15	97.27	95.20	74.79	71.76	75.73	77.59	85.83	86.40	94.95	95.37	96.96	97.74	97.58	**98.58**
AA(%)	87.05	80.23	86.96	88.61	94.38	90.97	90.86	90.11	97.33	96.69	97.08	95.80	81.84	79.51	82.81	84.39	88.13	88.95	94.98	95.84	97.11	97.95	98.09	**98.56**
KA(%)	83.69	70.22	81.29	82.68	92.14	87.11	87.75	85.20	95.49	94.95	96.42	93.73	68.32	64.76	69.60	71.74	81.69	82.48	93.40	93.96	96.02	97.04	96.83	**98.13**
Time(s)	719	109	66	23	546	17	26	**4**	560	47	40	17	1158	130	187	67	976	70	161	36	1047	157	200	69

Fig. 11 illustrates the results of the compared six categories of methods as a function of the number of training samples for the ITRES CASI-1500 UH data set. It can be seen that among the four tensorial methods, GTR provides competitive results for the first three kinds of tensorial features and gives the best results for the last three kinds of tensorial features. When compared with the multiple kernel-based methods, GTR outperforms MKL for the MK, MFK, and MSK features

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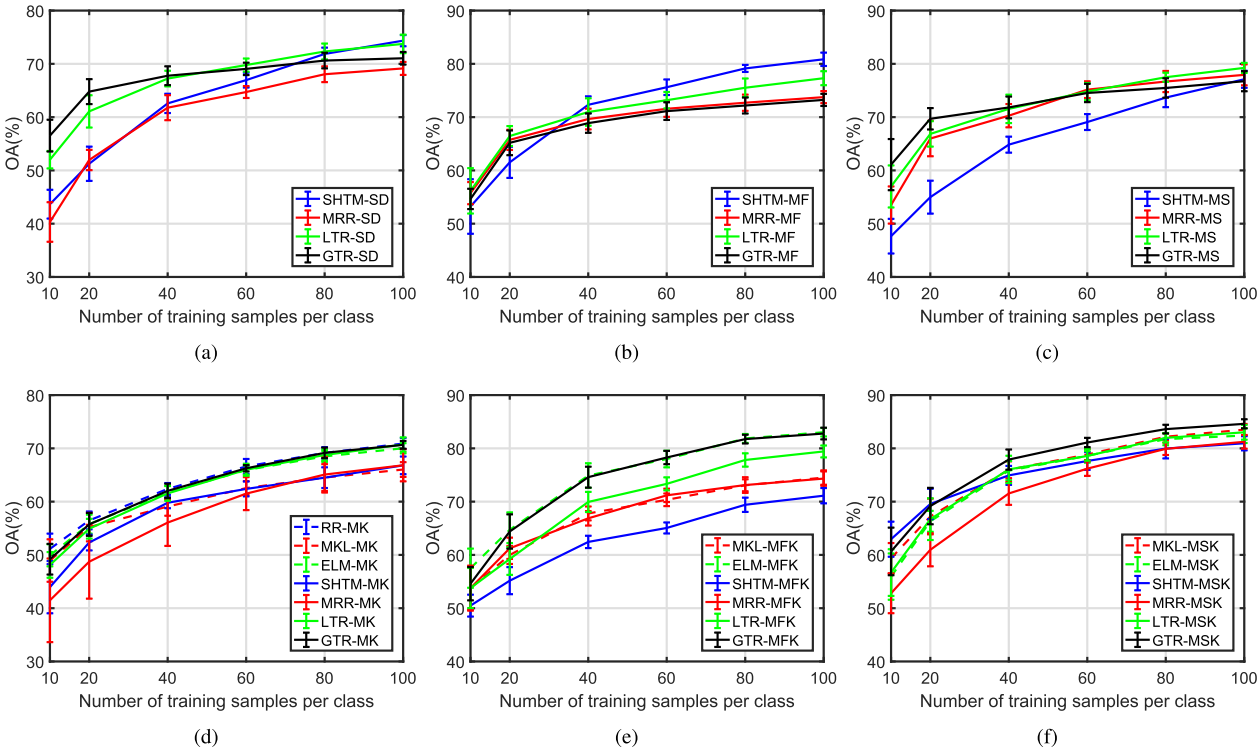


Fig. 11. OA as a function of the number of training samples for six tensorial scenes when applied to the ITRES CASI-1500 UH data set. (a) SD. (b) MF. (c) MS. (d) MK. (e) MFK. (f) MSK.

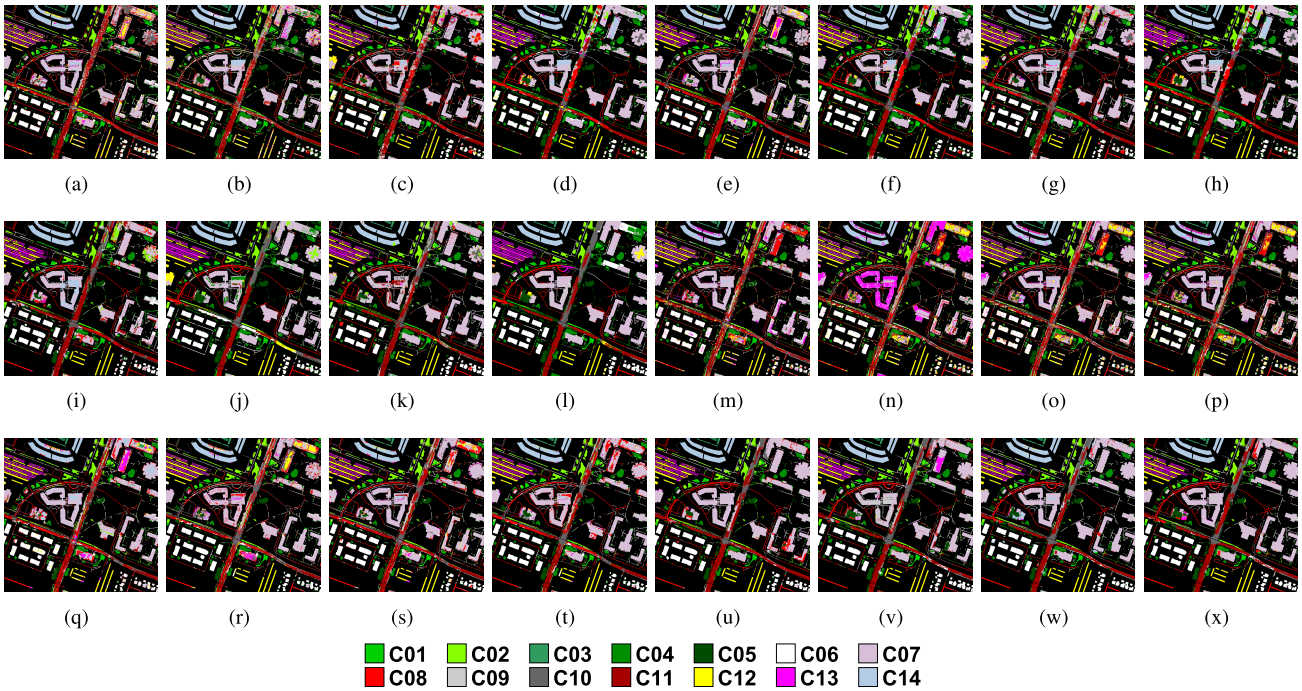


Fig. 12. ITRES CASI-1500 UH data set. (a) SHTM-SD. (b) MRR-SD. (c) LTR-SD. (d) GTR-SD. (e) SHTM-MF. (f) MRR-MF. (g) LTR-MF. (h) GTR-MF. (i) SHTM-MS. (j) MRR-MS. (k) LTR-MS. (l) GTR-MS. (m) SHTM-MK. (n) MRR-MK. (o) LTR-MK. (p) GTR-MK. (q) SHTM-MFK. (r) MRR-MFK. (s) LTR-MFK. (t) GTR-MFK. (u) SHTM-MSK. (v) MRR-MSK. (w) LTR-MSK. (x) GTR-MSK.

and outperforms ELM for the MSK features. For the four tensorial methods, Table VI reports their quality measures and processing times of the usage of 40 training samples per class. From this table, we can see that GTR gives the competitive results for the first three kinds of tensorial features and the best results for the last three kinds of tensorial features. Moreover, GTR reports the highest global accuracy for the MSK features. For the computation cost, it is clear that in all cases, GTR is the most economical one and SHTM is the most time-consuming one. Fig. 12 shows the classification maps

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TABLE VI
CLASSIFICATION ACCURACIES FOR THE ITRES CASI-1500 UH DATA SET USING DIFFERENT CLASSIFICATION METHODS

Class	SD				MF				MS				MK				MFK				MSK			
	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR	SHTM	MRR	LTR	GTR
C01	82.75	87.64	79.44	92.72	86.31	**93.21**	81.78	92.39	84.30	90.14	82.54	86.89	88.15	91.58	91.43	91.91	87.20	92.90	90.04	93.14	84.00	91.76	91.42	92.37
C02	77.79	70.23	72.09	73.35	83.94	74.55	72.88	72.90	77.10	79.66	72.42	79.77	81.44	78.89	81.15	81.55	82.15	85.05	82.64	84.32	82.17	83.36	**85.06**	84.99
C03	98.54	99.63	98.98	99.83	99.81	99.98	99.30	**100**	**100**	**100**	**100**	**100**	99.32	**100**	**100**	**100**	97.24	**100**	99.19	99.94	**100**	**100**	**100**	**100**
C04	86.35	89.91	88.36	92.95	89.23	94.04	90.44	94.26	84.26	92.90	87.10	**96.03**	87.59	88.73	88.80	89.06	84.28	92.63	88.99	94.21	85.97	88.12	88.65	91.25
C05	85.19	99.00	93.89	99.62	87.80	**99.63**	95.29	99.60	82.04	96.76	91.72	96.31	80.36	87.82	88.09	88.34	80.89	97.59	92.12	97.89	86.37	91.44	91.62	92.56
C06	82.37	81.32	83.42	86.01	89.62	88.51	88.04	87.04	94.26	90.28	93.68	95.31	75.20	82.60	79.55	82.81	82.90	90.77	89.77	93.13	**95.70**	92.53	94.58	95.10
C07	59.19	64.60	70.78	73.31	75.15	74.74	74.24	75.19	59.56	73.82	73.93	78.88	64.41	45.69	60.85	61.85	59.21	63.08	66.29	72.84	80.03	69.01	78.72	**81.36**
C08	46.76	26.93	45.52	39.91	45.98	41.30	51.19	40.82	54.78	50.42	58.49	52.90	32.95	37.12	37.68	37.27	43.43	41.57	52.18	55.84	60.81	63.37	65.76	**67.74**
C09	33.68	24.54	35.23	14.36	45.86	18.82	38.72	14.49	28.45	26.69	35.15	20.15	31.13	34.15	33.81	34.44	37.13	46.27	49.48	**50.84**	40.35	37.44	42.84	44.31
C10	57.70	44.89	64.14	61.21	78.04	64.02	67.52	66.53	79.73	72.90	77.13	63.75	61.75	67.67	66.80	67.73	63.87	77.46	74.02	**88.01**	73.26	85.47	81.42	86.74
C11	49.12	47.34	48.37	54.01	54.58	56.76	53.74	56.54	49.55	47.73	53.62	52.87	31.90	37.41	39.05	38.37	40.99	44.32	53.37	56.92	54.11	55.71	57.28	**59.38**
C12	76.68	65.41	77.54	74.12	84.29	78.37	84.54	68.20	86.04	91.68	88.12	61.86	79.25	86.14	85.34	86.43	81.43	87.57	89.69	93.38	87.77	91.33	91.71	**93.52**
C13	62.58	56.34	80.11	86.31	80.57	82.06	80.35	87.63	67.15	83.99	81.33	**94.03**	61.55	60.04	67.12	61.72	79.67	76.04	83.33	88.12	79.57	80.84	83.48	83.24
C14	90.13	94.41	92.19	97.11	96.17	98.30	94.70	99.10	99.43	99.75	96.34	**99.98**	80.94	83.26	84.99	84.16	92.71	91.06	90.51	97.76	99.97	98.38	97.69	99.63
OA(%)	62.58	61.77	67.24	67.77	72.32	69.64	70.97	68.88	64.84	70.27	71.55	71.82	59.79	56.07	61.54	62.02	62.44	66.87	69.94	74.60	74.93	71.54	76.05	**77.89**
AA(%)	70.63	68.01	73.58	74.63	78.38	76.02	76.62	75.33	74.76	78.34	77.97	77.05	68.28	70.08	71.76	71.83	72.36	77.59	78.69	83.31	79.29	80.63	82.16	**83.73**
KA(%)	57.22	56.25	62.11	62.76	67.79	64.85	66.26	63.95	59.78	65.68	66.99	67.28	53.79	50.63	56.09	56.55	56.92	62.12	65.37	70.57	70.57	67.05	72.05	**74.12**
Time(s)	2410	129	41	25	1832	22	35	**5**	1978	59	51	16	3905	480	1085	184	1999	260	914	101	4066	600	1106	169

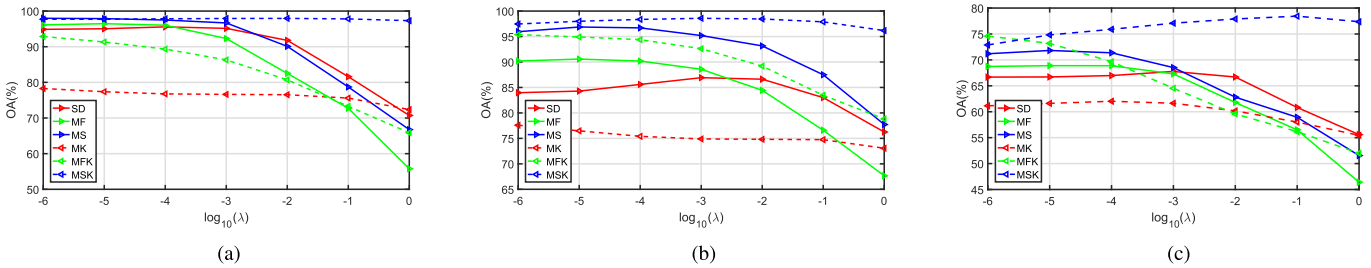


Fig. 13. Influence of parameter λ. (a) AVIRIS IndianP data set. (b) ROSIS PaviaU data set. (c) ITRES CASI-1500 UH data set.

associated with Table VI. Visually, these classification maps confirm the reported numerical results.

F. Influence of Parameter λ

In the proposed GTR method, there is only one parameter λ that is used to balance the fidelity and regularization terms. Although the parameter λ can be estimated by cross validation, we examine how it affects the classification performance on the given three data sets to provide some recommended settings for them. Here, 5% training samples per class are used for the AVIRIS IndianP data set and 40 for the ROSIS PaviaU data set and the ITRES CASI-1500 UH data set. Fig. 13 shows the impact of λ for the six instantiations described in Section III-C, where λ is varied from 10^{-6} to 10^{0}. It can be seen that there is a wide optimal range for the choice of λ.

V. CONCLUSION

This article has proposed a new GTR approach for the classification of hyperspectral images. The proposed approach is a tensorial extension of multivariate RR, and the CP decomposition is utilized to reduce the freedom of model and leverage the intrinsic structure of tensors. Inspired by the specific formulation of the model, the nonnegativity is exploited as the prior knowledge of all modes but the first one. Thus, the nonnegative constraint is imposed on the coefficient weights to promote their sparsity and further improve the discrimination of model. The model of the proposed approach is interpretable and can be degenerated into the multivariate RR. The proposed approach with rank-1 CP decomposition can be solved by alternating update directly, and for higher-rank CP decomposition, a divide-and-conquer strategy is used to simplify the model. The specific properties of the proposed approach provide it with the ability to deal with generalized tensors, and six kinds of tensorial features are constructed for investigation. The proposed approach has been experimentally tested using three different hyperspectral scenes, and the experimental results demonstrate that the proposed methods are not sensitive to initialization and exhibit good performance as compared to traditional tensor-based classifiers and multiple kernel learning methods. Although the results obtained by the proposed approach are very encouraging, further enhancements such as the additional prior knowledge of different modes and kernel tensor analysis should be pursued in future developments.

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Jianjun Liu (M'15) received the B.S. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence system from the Nanjing University of Science and Technology, Nanjing, China, in 2009 and 2014, respectively.

Since 2018, he has been a Post-Doctoral Researcher with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong. He is currently an Associate Professor with the School of Internet of Things Engineering, Jiangnan University, Wuxi, China. His research interests include hyperspectral image classification, superresolution, spectral unmixing, sparse representation, computer vision, and pattern recognition.



Zebin Wu (M'13–SM'17) received the B.S. and Ph.D. degrees in computer science from the Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2008, respectively.

He is currently a Professor with the School of Computer Science, Nanjing University of Science and Technology. His research interests include hyperspectral image processing, high-performance computing, and computer simulation.



Liang Xiao (M'11) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from the Nanjing University of Science and Technology, Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. He is currently a Professor with the School of Computer Science, Nanjing University of Science and Technology. His research interests include inverse problems in image processing, scientific computing, data mining, and pattern recognition.



Jun Sun received the Ph.D. degree in control theory and engineering and the M.Sc. degree in computer science and technology from Jiangnan University, Wuxi, China, in 2009 and 2003, respectively.

He is currently a Full Professor with the Department of Computer Science and Technology, Jiangnan University, where he is also the Vice Director of the Jiangsu Provincial Engineering Laboratory of Pattern Recognition and Computational Intelligence. His research interests include computational intelligence, machine learning, bioinformatics, among others. He has authored or coauthored more than 150 articles in journals, conference proceedings, and several books in the above areas.



Hong Yan (M'89–SM'93–F'06) received the B.S. degree in electrical engineering from the Nanjing University of Posts and Telecommunications, Nanjing, China, in 1982, the M.S. degree in electrical engineering from the University of Michigan, Ann Arbor, MI, USA, in 1984, and the Ph.D. degree in electrical engineering from Yale University, New Haven, CT, USA, in 1989.

From 1986 to 1989, he was a Research Scientist with General Network Corporation, New Haven, where he researched on design and optimization of computer and telecommunications networks. In 1989, he joined the University of Sydney, Sydney, NSW, Australia, where he became a Professor of imaging science in 1997. He is currently a Professor of electrical engineering with the City University of Hong Kong, Hong Kong. His research interests include image processing, pattern recognition, and bioinformatics. He has more than 600 publications in these areas.

Dr. Yan was elected an IAPR Fellow for contributions to document image analysis and an IEEE Fellow for contributions to image recognition techniques and applications. He was a recipient of the 2016 Norbert Wiener Award from the IEEE SMC Society for contributions to image and biomolecular pattern recognition techniques.

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